Telco Customer Churn Prediction Using Machine Learning

Introduction

Customer churn is a major challenge for telecommunication companies because losing customers directly impacts revenue and business growth. The purpose of this project is to analyze customer behavior and predict whether a customer is likely to leave the company using machine learning techniques.

Objective

The main objectives of this project are:

- Analyze customer data and identify churn patterns.
- Build machine learning models to predict customer churn.
- Segment customers based on their value and risk level.
- Identify high-risk customers for retention strategies.

Dataset

The project uses the Telco Customer Churn dataset. The dataset contains customer demographic information, service details, contract information, billing details, and churn status.

Important attributes include:

1. Tenure
2. Monthly Charges
3. Total Charges
4. Contract Type
5. Internet Service
6. Payment Method
7. Churn

Methodology

Data Preprocessing

The following preprocessing steps were performed:

- Removed unnecessary spaces from column names.
- Converted TotalCharges to numeric format.
- Replaced missing values using the median.
- Removed duplicate records.

Feature Engineering

Additional features were created:

- Average Spend Per Month
- Customer Lifetime Value
- Long-Term Customer Indicator

Exploratory Data Analysis

Several visualizations were generated:

- Churn Distribution
- Revenue Trend Analysis
- Correlation Heatmap
- Customer Segmentation Scatter Plot

Customer Segmentation

Customers were categorized into:

- High Value Customers
- Medium Value Customers
- Low Value Customers
- based on their spending behavior.

Machine Learning Models

Two machine learning algorithms were implemented:

1. Logistic Regression
2. Random Forest Classifier

Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix

Risk Analysis

Customers were classified into:

- High Risk

- Medium Risk
- Low Risk

based on churn probability predictions generated by the Random Forest model.

Results

The Random Forest model provided strong predictive performance and successfully identified customers who are likely to churn. Feature importance analysis showed that factors such as tenure, contract type, monthly charges, and total charges have significant influence on churn behavior.

Conclusion

This project demonstrates how machine learning can be used to predict customer churn and support customer retention strategies. The developed system helps telecom companies identify valuable customers, detect churn risks, and make data-driven business decisions.